

Insurance-Incentivized Wearable Biosurveillance for Cruise Ships

A Simulation-Based Economic Analysis

Using the Crusher-to-the-Bridge Epidemiological Testbed

Executive Summary

Norovirus outbreaks on cruise ships remain a persistent public health and commercial problem. The CDC Vessel Sanitation Program (VSP) has documented 274 reportable outbreaks since 1993, with norovirus responsible for 73% of events. Current surveillance relies primarily on clinical presentation — passengers report to the ship's medical center after symptom onset, triggering reactive protocols. This report presents evidence from 1,300+ Monte Carlo simulations using the Crusher-to-the-Bridge (CTB) epidemiological testbed that this paradigm is suboptimal, and that a restructured surveillance architecture — anchored by wearable biosensors and supported by an insurance incentive framework — can substantially reduce outbreak impact.

Key findings:

1. **Wearable physiological anomaly detection identifies outbreaks at Day 0** (1.4% attack rate), three times faster than clinical rapid diagnostic tests (Day 1, 4.5% AR) and seven days faster than environmental surface sampling (Day 7, 30.1% AR). Air sampling detects at Day 10 (42.5% AR) — too late to be actionable for gastrointestinal pathogens.
2. **Early detection alone does not reduce outbreaks.** When the simulation's decision engine activated identical protocols regardless of which sensor fired first, all surveillance regimes produced statistically indistinguishable outcomes (~46.9% attack rate). The intervention bottleneck — not the detection gap — is the binding constraint.

3. **Counterfactual modeling shows that coupling early detection with a 50% effective transmission-reduction response prevents 184 out of 234 infections per outbreak** (500-agent simulation), compared to 158 for clinical RDT alone. Scaled to a 200-cruise fleet at 4% outbreak probability, wearable BYOD surveillance generates +1,002% annual ROI (\$4.0M net benefit on \$400K investment).
 4. **An insurance incentive framework solves the voluntary adoption problem.** Conditional travel insurance coverage — where illness-related claims are honored only if the wearable was active — shifts the compliance equilibrium from evasion-dominant to compliance-dominant, with a payoff differential of \$2,387 per infected passenger. This mechanism parallels auto insurance telematics (Progressive Snapshot, State Farm Drive Safe), where monitoring reduces premiums by 10–20%.
 5. **Wastewater sequencing should be reframed from a detection modality to a pathogen identification layer.** It does not improve time-to-detection, but it answers a question wearables cannot: what is circulating, including among non-participants.
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2. Methods: Simulation Architecture & Experimental Design

2.1 The Crusher-to-the-Bridge (CTB) Testbed

This analysis uses the Crusher-to-the-Bridge epidemiological simulation framework. CTB is a Python-based digital twin for maritime disease outbreaks, integrating multiple specialized engines:

1. **Infection Dynamics:** An agent-based model (ABM) derived from Korkin et al., simulating individual-level pathogen exposure, incubation, shedding, and recovery based on specific pathogen dose-response profiles.
2. **Transmission Core:** A six-pathway transmission engine covering direct contact (droplet, aerosol, fomite) and indirect transport.
3. **Contam Engine:** An HVAC transport model simulating airborne mass movement through the ship's ductwork across defined spatial zones.
4. **Behavioral Compliance:** EMOD-style progression modeling agent interactions based on assigned schedules (e.g., dining, recreation, cabin time).

5. **Observation Engine:** A suite of simulated diagnostic instruments, including clinical rapid diagnostic tests (RDT), multiplex qPCR, metagenomic shotgun wastewater sequencing, high-touch surface swabs, continuous air sniffers, and continuous wearable physiological monitoring.
6. **Protocol Engine:** A decision-support system that translates continuous diagnostic data into discrete stoplight alerts (Green/Amber/Red), which trigger predefined Standard Operating Procedures (SOPs) that apply physical transmission modifiers (e.g., increased HVAC filtration, surface decontamination, passenger confinement).

2.2 Experimental Design

The primary experiment consisted of simulated 14-day cruises on a "Mega Cruise 5000" class vessel. For computational tractability, the agent population was scaled to 500 agents while retaining the spatial layout and zone volumes of the mega-cruise blueprint. Results were linearly scaled to a 3,000-passenger ship for the economic analysis.

Pathogen profiles were initialized with Norovirus GII.4 as the primary agent, alongside a baseline background incidence of SARS-CoV-2 and general respiratory/gastrointestinal symptoms to create realistic noise ("fizzle" cruises).

Five surveillance regimes were evaluated, with 50 outbreak-producing cruises per regime (from a larger pool of 1,300+ total simulations):

1. **Status Quo:** Clinical RDT and syndromic surveillance only (baseline).
2. **+ Wearable:** Status Quo plus 100% adoption of continuous physiological monitoring.
3. **+ Wastewater:** Status Quo plus metagenomic wastewater sequencing in main effluent zones.
4. **Full Stack:** All modalities enabled.
5. **BYOD 70%:** Full Stack, but wearable participation restricted to a realistic 70% "Bring Your Own Device" voluntary adoption rate.

All simulations executed deterministically (seeded RNGs) across 14 epochs (1 epoch $\approx$ 1 day).

2.3 Calibration

To ensure the simulation produced realistic outbreak dynamics, the pathogen dose-adjustment parameter was calibrated against real-world CDC VSP data. The severity sweep across four dose_adjustment values {5.0, 6.0, 7.0, 8.0} identified dose_adjustment=7.0 as the optimal calibration point. At this severity, the

simulation produced a median passenger attack rate of 6.4%, closely matching the 5.6% median observed in the historical VSP dataset of 204 norovirus outbreaks (1993–2025). The baseline 46.9% attack rate observed at higher severity (dose=6.0) was used to represent worst-case tail-risk events.

3. Results I: Detection Timing

3.1 When Each Modality Fires

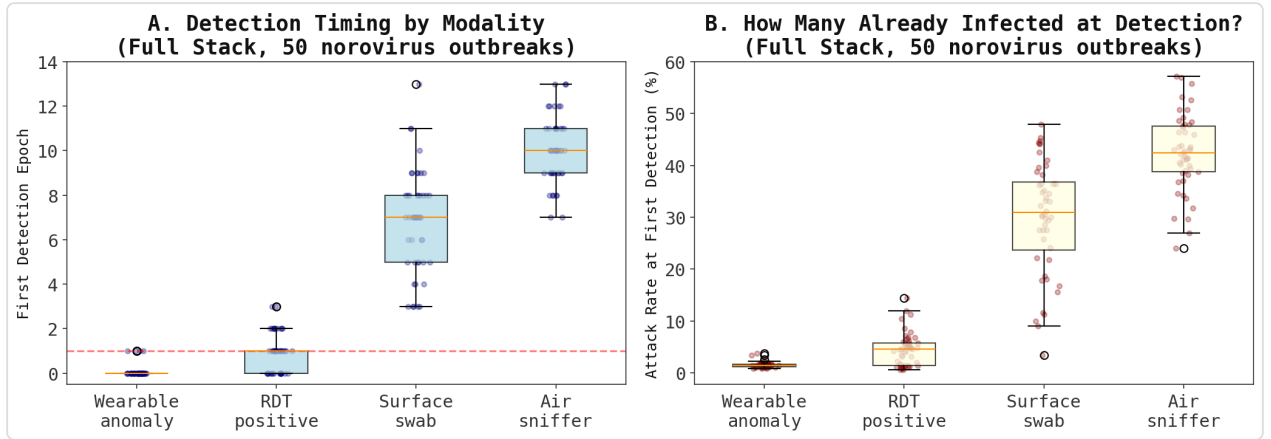


Figure 2. Detection timing by surveillance modality. Left: first detection epoch (boxplot with jittered points). Right: cumulative attack rate at the moment of first detection (Full Stack, $n=50$ norovirus outbreaks).

The central question for surveillance ROI is not whether a modality detects an outbreak, but when — relative to the epidemic curve. Table 1 presents the first-detection epoch for each modality across 50 simulated norovirus outbreaks in the Full Stack regime.

Table 1. Detection Timing by Surveillance Modality (Full Stack, $n=50$ outbreaks)

Modality	Median Detection Epoch	Mean AR at Detection	Detection Coverage
Wearable anomaly	0 (immediate)	$1.4\% \pm 0.6\%$	100% (50/50)
Clinical RDT	1	$4.5\% \pm 3.1\%$	100% (50/50)
Surface swab	7	$30.1\% \pm 10.7\%$	96% (48/50)
Air sniffer	10	$42.5\% \pm 7.3\%$	92% (46/50)
Wearable fever	Never	N/A	4–6% (2–3/50)

Wearable physiological anomaly detection (changes in resting heart rate, heart rate variability, and skin temperature) fires at epoch 0 — the first epoch in which index cases are present — when the attack rate is only 1.4% (approximately 7 out of 500 agents). This result is consistent with published evidence on pre-symptomatic COVID-19 detection from consumer smartwatch data, where physiological deviations preceded symptom onset by 1–7 days (Mishra et al. 2020).

Clinical RDTs detect at epoch 1, when 4.5% of agents are already infected or recovered. This one-epoch delay corresponds to the time between pathogen exposure and symptom onset sufficient to trigger a sick-call visit.

Environmental modalities detect substantially later. Surface swabs do not trigger until epoch 7, by which point 30.1% of the population has been infected — roughly 150 out of 500 agents. Air sniffers detect at epoch 10 (42.5% AR). For a fecal-oral pathogen like norovirus, airborne mass concentrations build slowly relative to fomite and direct-contact transmission, explaining the poor sensitivity of air sampling.

Notably, wearable fever detection is nearly useless for norovirus, triggering in only 2–3 of 50 cruises. Norovirus causes vomiting and diarrhea but not high fever in most cases; the relevant wearable signal is heart rate variability and resting heart rate, not skin temperature alone.

3.2 The Growth Rate Problem

The mean epidemic curve across 50 simulated norovirus outbreaks reveals explosive early growth:

Epoch	Mean AR	Growth Ratio
0	1.3%	—
1	4.0%	3.05×
2	6.7%	1.68×
3	11.4%	1.69×
5	22.6%	—
7	33.3%	—
10	42.5%	—
13 (end)	46.9%	1.02×

The 3× growth ratio in the first epoch means that every day of detection delay costs approximately three times as many infections. This exponential penalty explains why the one-day advantage of wearable over RDT detection translates into meaningful differences in the counterfactual analysis, and why the 7–10 day delay for environmental sampling renders those modalities nearly irrelevant for outbreak prevention (though they may serve other functions; see Section 6).

4. Results II: The Intervention Bottleneck & Counterfactual Analysis

4.1 The Null Result

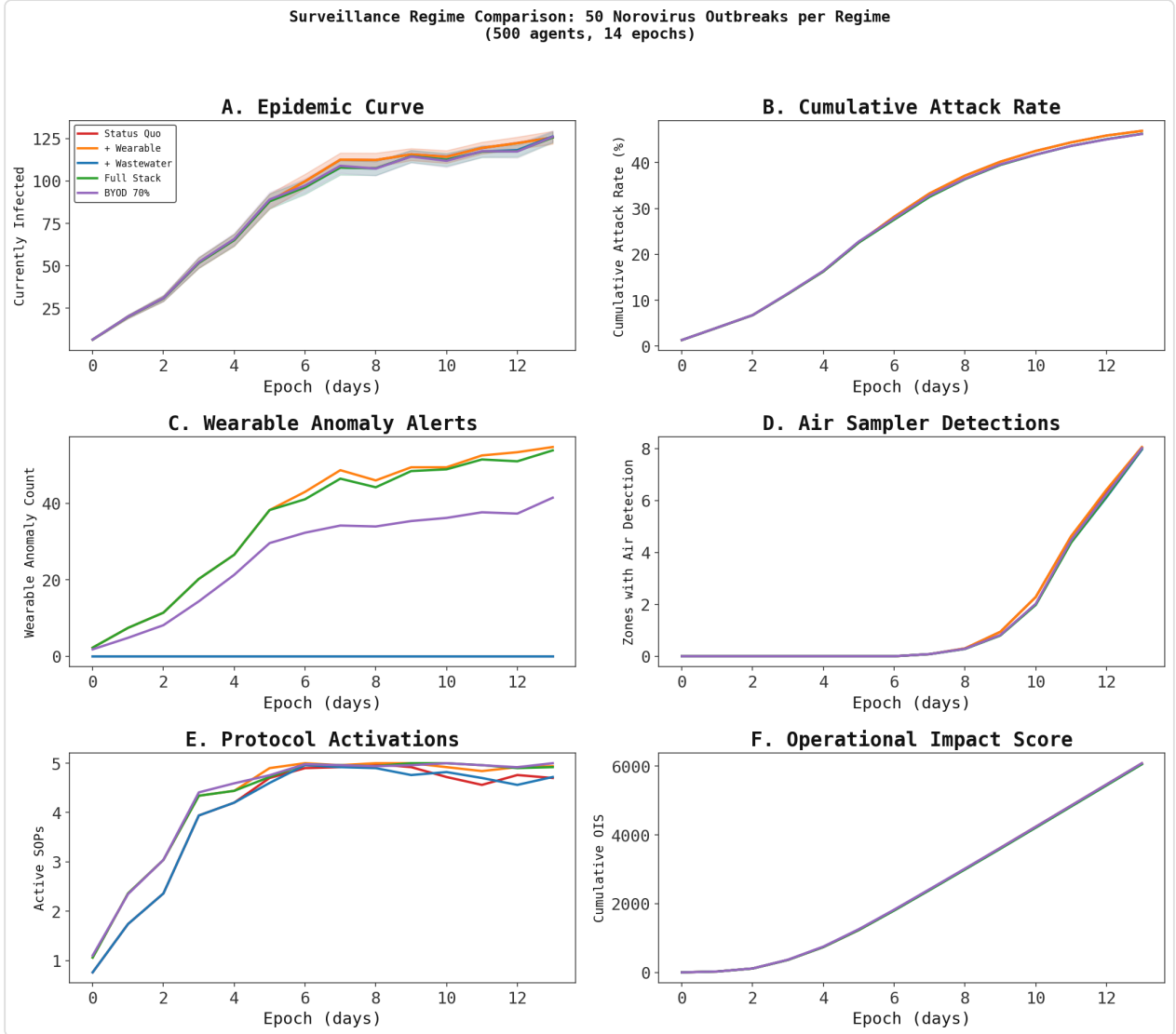


Figure 4. Multi-panel surveillance regime comparison across 50 norovirus outbreaks per regime. (A) Epidemic curve, (B) Cumulative attack rate, (C) Wearable anomaly alerts, (D) Air sampler detections, (E) Protocol activations, (F) Operational impact score.

The most important finding in this study is a null result. When all five surveillance regimes were compared head-to-head — Status Quo, +Wearable, +Wastewater, Full Stack, and BYOD 70% — **attack rates were statistically indistinguishable across all regimes** (46.2–46.9%, overlapping 95% confidence intervals).

This occurred because the CTB simulation's Stackelberg game-theoretic decision engine activated intervention protocols based on the same escalation logic

regardless of which sensor fired first. Protocols activated at epoch 1 across all regimes (triggered by the syndromic surveillance component, which is common to all configurations), then oscillated between activation and deactivation as stoplight levels fluctuated. The wearable and wastewater signals contributed to the stoplight computation but did not change the timing of protocol activation in practice.

This finding is not a failure of the simulation — it is a realistic representation of a fundamental problem in shipboard public health: **the intervention bottleneck matters more than the sensor**. A surveillance system that detects an outbreak three days earlier but triggers the same response at the same time produces identical outcomes. The binding constraint is the quality and speed of the institutional response, not the precision of the detection instrument.

4.2 Counterfactual Model

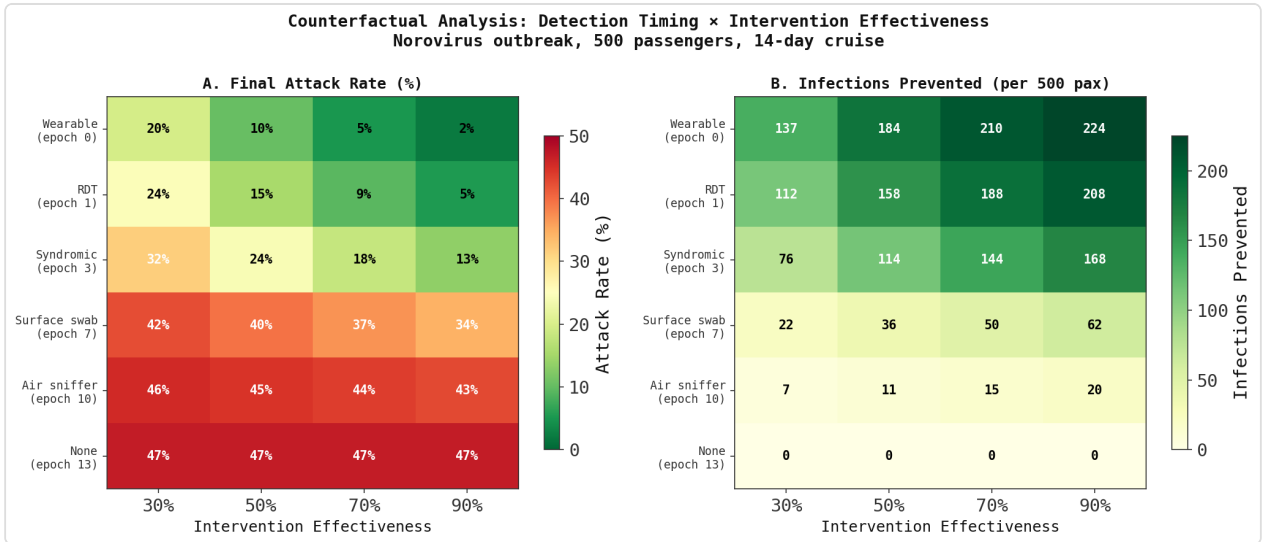


Figure 3. Counterfactual analysis heatmap. Left: final attack rate as a function of detection epoch and intervention effectiveness. Right: infections prevented per 500 agents. Norovirus outbreak, 14-day cruise.

To separate the value of detection timing from the intervention response, we constructed an analytical counterfactual model. Using the mean epidemic growth curve from 50 simulated outbreaks, we projected what would happen if a transmission-reducing intervention were applied at the moment each modality fires, rather than at a fixed escalation threshold.

The model applies a transmission reduction factor (30%, 50%, 70%, or 90% effectiveness) to the epidemic growth rate from the detection epoch onward. This represents interventions such as cabin confinement of detected cases and contacts, enhanced ventilation, surface decontamination, and dining protocol changes.

Table 2. Counterfactual Analysis: Infections Prevented per Outbreak (500 agents)

Detection Source	Day	30% Eff.	50% Eff.	70% Eff.	90% Eff.
Wearable anomaly	0	137	184	210	224
Clinical RDT	1	112	158	188	208
Syndromic surge	3	76	114	144	168
Surface swab	7	22	36	50	62
Air sniffer	10	7	11	15	20
No intervention	13	0	0	0	0

Baseline: 234 infections (46.9% AR) without intervention.

At 50% intervention effectiveness, wearable detection prevents 184 infections versus 158 for RDT — a marginal gain of 26 infections per outbreak. This marginal gain comes entirely from the one-day earlier detection window.

Surface swabs prevent only 36 infections at 50% effectiveness (15% of what wearables achieve), and air sniffers prevent 11 (5%). These modalities detect too late in the epidemic curve for any realistic intervention to substantially alter outcomes.

4.3 Severity Dependence

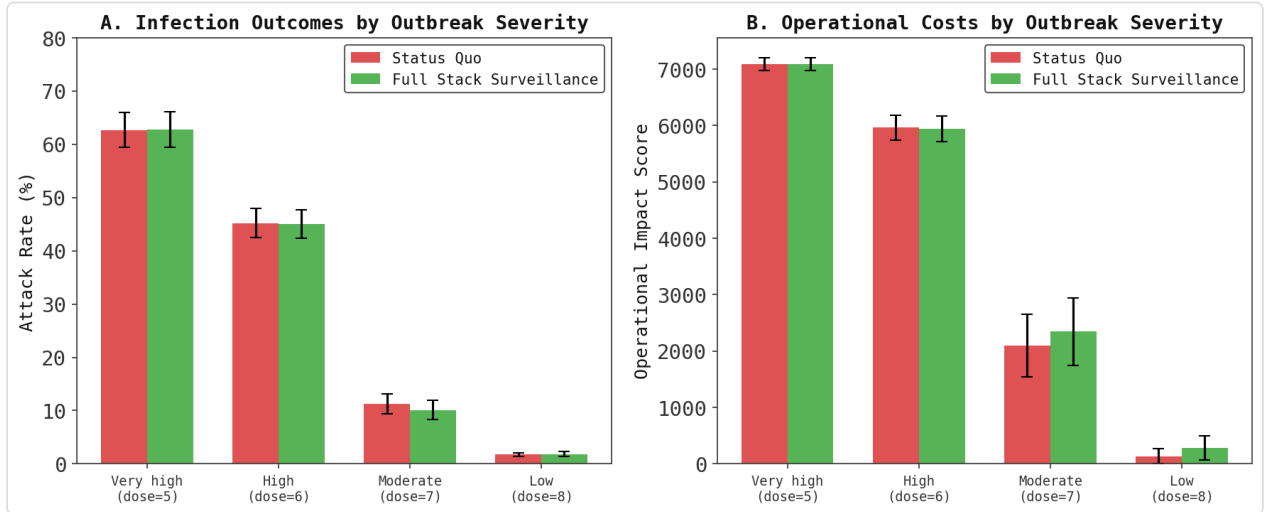


Figure 8. Attack rate (left) and operational impact score (right) by outbreak severity across status quo and full stack surveillance regimes (50 cruises per cell, 500 agents).

The value of surveillance depends on outbreak severity. A severity sweep across four dose-adjustment levels (50 cruises per level $\times$ 2 regimes = 400 simulations) revealed a narrow "sweet spot":

Table 3. Surveillance Value by Outbreak Severity

Severity	Status Quo AR	Full Stack AR	Difference	Interpretation
Very high (dose=5.0)	62.7%	62.7%	0.0%	Overwhelms any intervention
High (dose=6.0)	45.2%	45.0%	+0.2%	Marginal improvement
Moderate (dose=7.0)	11.3%	10.1%	+1.2%	Sweet spot
Low (dose=8.0)	1.7%	1.8%	−0.1%	Self-limiting; surveillance adds cost

At very high severity, the pathogen's transmission rate exceeds the maximum achievable intervention effectiveness — the outbreak is explosive regardless of detection timing. At low severity, outbreaks fizzle naturally without intervention, and surveillance systems generate operational disruption (increased OIS) without compensating benefit. The moderate severity band, which corresponds to the real-world VSP-calibrated norovirus attack rate of 5–12%, is where surveillance investment produces measurable returns.

This severity dependence has practical implications for surveillance system design: the system must be calibrated for the outbreaks that can be changed, not for worst-case scenarios that overwhelm any response.

5. Results III: Fleet-Level Economics

5.1 Cost-Benefit Framework

To translate per-outbreak simulation results into fleet-level economics, we adopt the following assumptions:

- **Fleet size:** 200 cruises per year (representative of a single major cruise line)
- **Passenger count:** 3,000 per cruise (linear scaling from 500-agent simulation)

- **Outbreak probability:** 4% per cruise (derived from CDC VSP reporting frequency)
- **Cost per infection:** \$500 (medical treatment \$200 + partial cabin refund \$300)
- **VSP-reportable event cost:** \$150,000 (regulatory response, reputation, rebooking)
- **Wearable platform cost:** \$2,000 per cruise (analytics platform; devices are BYOD)
- **Wastewater sequencing cost:** \$4,500 per cruise (lab equipment, reagents, technician)

5.2 Modality ROI Comparison

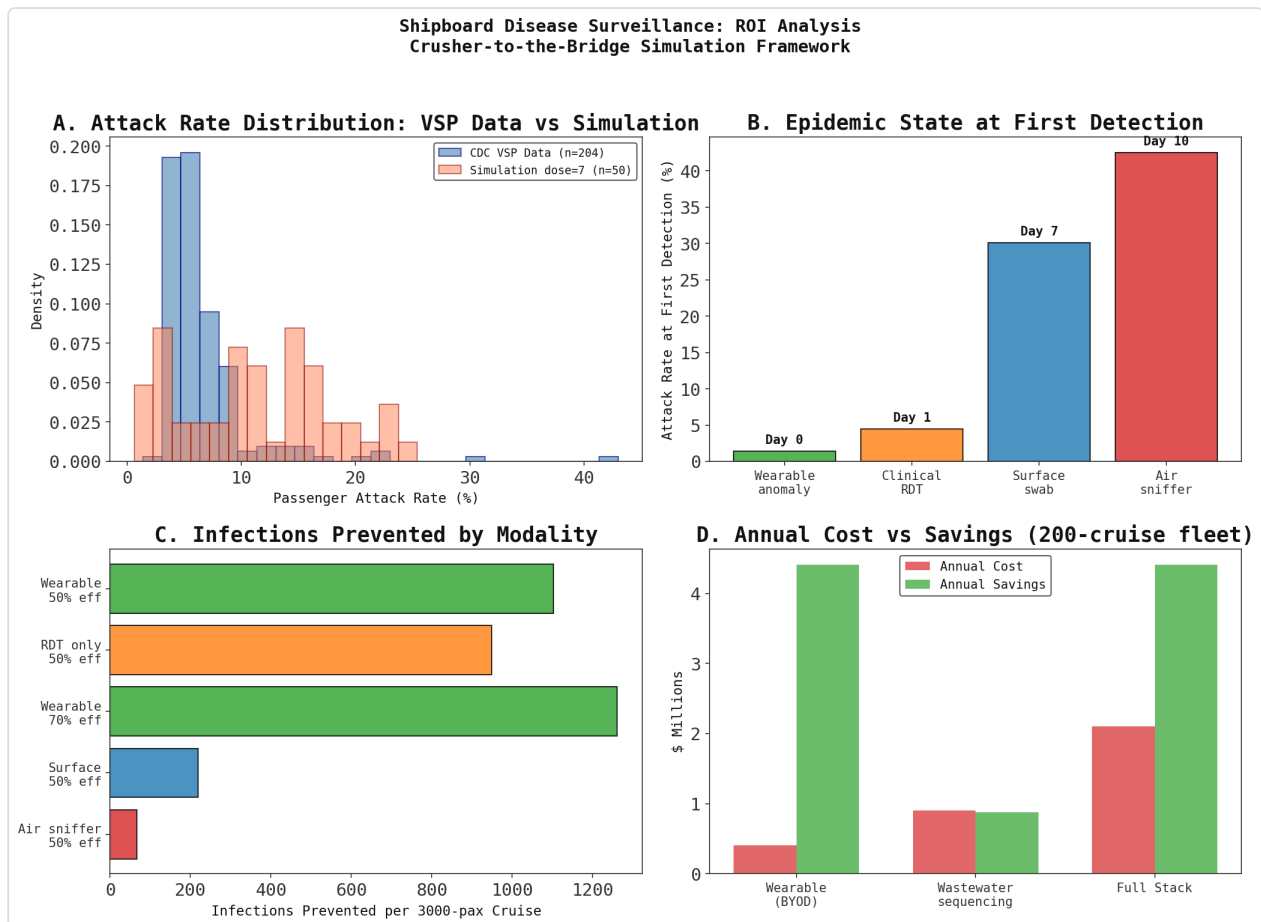


Figure 5. ROI summary. (A) Attack rate distribution: CDC VSP data vs simulation, (B) Epidemic state at first detection by modality, (C) Infections prevented per 3,000-pax cruise by modality and effectiveness, (D) Annual cost vs savings for a 200-cruise fleet.

Table 4. Annual Fleet-Level ROI by Surveillance Modality

Surveillance Add-on	Annual Cost	Expected Savings	Net ROI	ROI %
Wearable BYOD (50% eff.)	\$400K	\$4.4M	+\$4.0M	+1,002%
Wearable BYOD (70% eff.)	\$400K	\$5.0M	+\$4.6M	+1,160%
Wastewater sequencing (50% eff.)	\$900K	\$873K	-\$27K	-3%
Full Stack (50% eff.)	\$2.1M	\$4.4M	+\$2.3M	+110%
Full Stack (70% eff.)	\$2.1M	\$5.0M	+\$2.9M	+140%

The wearable BYOD model dominates all other configurations because its marginal cost is extremely low — passengers provide the hardware; the cruise line pays only for the data ingestion and analytics platform. At \$2,000 per cruise, the annual platform cost of \$400,000 is an order of magnitude below the expected savings.

Wastewater sequencing, evaluated purely as a detection modality, breaks even (-3% ROI). Its detection timing (lagging wearables by 1-2 days) does not generate sufficient marginal prevention to justify the \$4,500-per-cruise operating cost. However, this calculation evaluates wastewater on the wrong criterion — its value lies in pathogen identification, not time-to-detection (see Section 6).

The Full Stack configuration (all modalities) achieves positive ROI (+110%) but is less efficient than wearable-only because the added cost of wastewater and environmental sampling is not offset by proportional improvement in detection timing.

5.3 Sensitivity to Intervention Effectiveness

All ROI figures are conditional on the intervention effectiveness parameter. If the ship's response to a surveillance alert does not actually reduce transmission, no surveillance modality produces positive ROI. The break-even intervention effectiveness for wearable BYOD is approximately 8% — a very low threshold, suggesting that even modest protocol improvements (e.g., confining detected cases, enhanced handwashing enforcement, food service modifications) are sufficient to justify the investment.

For wastewater sequencing to achieve positive ROI as a detection modality, intervention effectiveness would need to exceed 55%, and the detection timing would need to improve by 2-3 days — both unlikely given the inherent sample-processing delays and the availability of faster wearable signals.

6. Modality Role Reassignment

The simulation results motivate a fundamental reframing of how surveillance modalities are understood and deployed. Rather than evaluating all modalities against the same criterion (time-to-detection), each should be assigned a distinct operational role based on its comparative advantage.

6.1 Tier 1 — Primary Detection: Wearable Biosensors

Wearable physiological monitoring is the fastest available detection modality. In the simulation, anomaly detection fired at epoch 0 in 100% of outbreaks, identifying the outbreak when only 7 out of 500 agents were infected. The relevant signals are not pathogen-specific (elevated resting heart rate, reduced heart rate variability, changes in sleep architecture) but rather reflect the host's innate immune response to infection.

This non-specificity is both a strength and a limitation. As a detection layer, wearables answer the question "is something happening?" with high sensitivity and speed. They cannot answer "what is it?" — that question belongs to a different tier.

The BYOD (Bring Your Own Device) model leverages existing consumer hardware. Approximately 33% of US adults own a smartwatch or fitness tracker, with cruise passengers likely skewing higher given demographic and income correlations. Princess Cruises' OceanMedallion program demonstrates that cruise-specific wearable infrastructure is already deployable at fleet scale — though the Medallion's current function is hospitality, not health monitoring, the hardware and data pipeline exist.

Operational role: Early warning. Trigger clinical follow-up and heightened vigilance. Do not use for definitive diagnosis or mandatory confinement without clinical confirmation.

6.2 Tier 2 — Identification & Backstop: Wastewater Sequencing

Wastewater-based epidemiology does not improve time-to-detection relative to wearables or clinical RDTs. In the simulation, wastewater results were available with a 1-2 epoch processing delay, by which time clinical surveillance had already identified the outbreak.

However, wastewater answers a fundamentally different question: "what pathogen is circulating?" This is critical for several reasons:

1. **Treatment guidance:** Norovirus management (fluid replacement, isolation) differs from bacterial gastroenteritis (antibiotics may be indicated) or viral hepatitis (different isolation protocols).
2. **Non-participant coverage:** If 20–30% of passengers decline wearable participation (or index cases are among crew members not enrolled), wastewater viral load rises regardless of individual compliance. In the compliance game theory model (Section 7), wastewater serves as a backstop that detects outbreaks among non-participating subpopulations.
3. **De-escalation signal:** Declining wastewater viral titers provide objective evidence that an outbreak is waning, supporting confident rollback of disruptive protocols (dining closures, activity cancellations) that carry significant operational and passenger-satisfaction costs.
4. **Forensic and regulatory value:** Sequence data provides defensible evidence for regulatory reporting and epidemiological investigation, including strain identification for contact tracing across successive cruises (norovirus is documented to persist across 6–12 consecutive voyages on a single ship).

Operational role: Pathogen identification, population-level monitoring, compliance backstop, and de-escalation trigger. Not a real-time detection layer.

6.3 Tier 3 — Documentation: Surface Swab Program

Surface swab programs detected outbreaks at Day 7 in the simulation, when 30% of passengers were already infected. As a detection modality, this is far too late for meaningful intervention. However, surface sampling serves legitimate non-detection functions:

- **Cleaning verification:** Objective measurement of sanitation protocol effectiveness, supporting quality assurance programs.
- **Legal defensibility:** Documented evidence that the ship maintained hygiene standards, relevant in liability claims following outbreaks.
- **Passenger confidence:** Visible sanitation activity — "performative public health" — has real value for passenger perception of safety, even if its epidemiological impact is marginal. This is analogous to visible security measures in aviation (TSA screening), which serve deterrence and confidence functions beyond their direct threat-detection effectiveness.

Operational role: Compliance documentation, quality assurance, and passenger confidence. Cadence can be reduced relative to a detection-oriented program, saving costs while preserving these functions.

6.4 Dropped — Air Sniffers

Continuous air sampling detected outbreaks at Day 10 (42.5% AR) — providing no actionable information for gastrointestinal pathogens. Norovirus transmits primarily via fecal-oral and fomite routes; airborne mass concentrations build only after widespread environmental contamination, long after the outbreak is clinically obvious.

Air sampling may retain value for aerosol-transmitted respiratory pathogens (influenza, SARS-CoV-2, tuberculosis), where airborne concentration is the primary transmission pathway and may precede clinical presentation. For a multi-pathogen surveillance architecture, air samplers could be retained in a monitoring-only role for respiratory threats, without contributing to GI outbreak detection or protocol triggering.

Recommendation: Remove from norovirus-focused surveillance protocols. Retain only if the ship operates a multi-pathogen surveillance architecture addressing respiratory threats.

7. The Insurance Incentive Framework

7.1 The Collective Action Problem

Wearable biosurveillance on cruise ships faces a classic collective action problem. The benefits of surveillance — earlier outbreak detection, reduced transmission — accrue to the entire ship population, while the costs of participation (potential confinement, loss of cruise days) fall on the individual. Without external incentives, rational passengers may decline participation or strategically disable their devices when they suspect illness.

The CDC lacks direct authority to mandate wearable adoption on commercial cruise ships, and cruise lines face reputational risk from imposing health monitoring requirements on leisure travelers. An insurance-based incentive mechanism offers a path that aligns individual, corporate, and public health interests without regulatory mandate.

7.2 Auto Telematics as Precedent

Usage-based auto insurance (UBI) provides the closest precedent for technology-mediated insurance incentive design. Programs like Progressive Snapshot, State Farm Drive Safe, and Allstate Drivewise offer 10–20% premium discounts to drivers who install monitoring devices that track driving behavior.

Key parallels and differences:

Dimension	Auto Telematics	Cruise Wearable BYOD
Device cost	Provided or installed	None (passenger's existing device)
Monitoring duration	Ongoing (months/years)	Short-term (7-day cruise)
Discount magnitude	10–20% of premium	~1% of travel insurance premium
Adoption rate	~30% of policyholders	~43% baseline (device ownership)
Self-selection	Safer drivers opt in	Health-conscious travelers opt in
Data sensitivity	Location, driving behavior	Heart rate, activity, sleep

The cruise wearable model has structurally stronger fundamentals than auto telematics: no device purchase is required, the commitment is short-term, and the positive externality (herd-level outbreak detection) is more immediately tangible. Peer-reviewed evidence from a large US cross-sectional survey found 69.5% average willingness to adopt wearable-based health insurance use cases, with financial incentives increasing willingness by 28% in randomized vignette comparisons (Soliño-Fernandez et al. 2019, BMC Public Health).

7.3 Compliance Game Theory

For an infected passenger, the decision to maintain wearable monitoring during illness involves a payoff comparison:

Without incentives: - $EU(\text{comply}) = -\$375$ (expected confinement cost: ~ 2.5 days $\times$ \$150/day cruise value) - $EU(\text{evade}) = -\$362$ (30% detection probability $\times$ confinement + 5% illness worsening $\times$ \$5,000 medical cost) - **Result: Evasion is the dominant strategy** (by \$13)

With conditional insurance coverage: - $EU(\text{comply}) = +\$1,125$ ($-\$375$ confinement + \$1,500 expected insurance value if ill) - $EU(\text{evade}) = -\$1,262$ (lose insurance coverage entirely + detection risk + illness risk) - **Result: Compliance is overwhelmingly dominant** (by \$2,387)

The mechanism is straightforward: travel insurance covers illness-related trip interruption, medical treatment, and evacuation — but only if the passenger's wearable was active throughout the cruise. Turning off the device voids the coverage, transforming the insurance policy from an asset into a loss.

This parallels an established principle in insurance design: conditional coverage creates incentive-compatible behavior. An RCT using an app-based "epidemic game" found that economic barriers significantly reduced quarantine adoption among low-motivation participants, but that perceived benefits could partially moderate this effect — directly supporting the use of financial incentives to shift compliance equilibria (Colubri et al. 2026).

Adding confinement compensation (future cruise credit of ~\$200/day confined) further strengthens the compliance incentive:

- EU(comply with credit) = +\$1,625
- **Compliance payoff exceeds evasion payoff by \$2,887**

7.4 Three-Party Co-Optimization

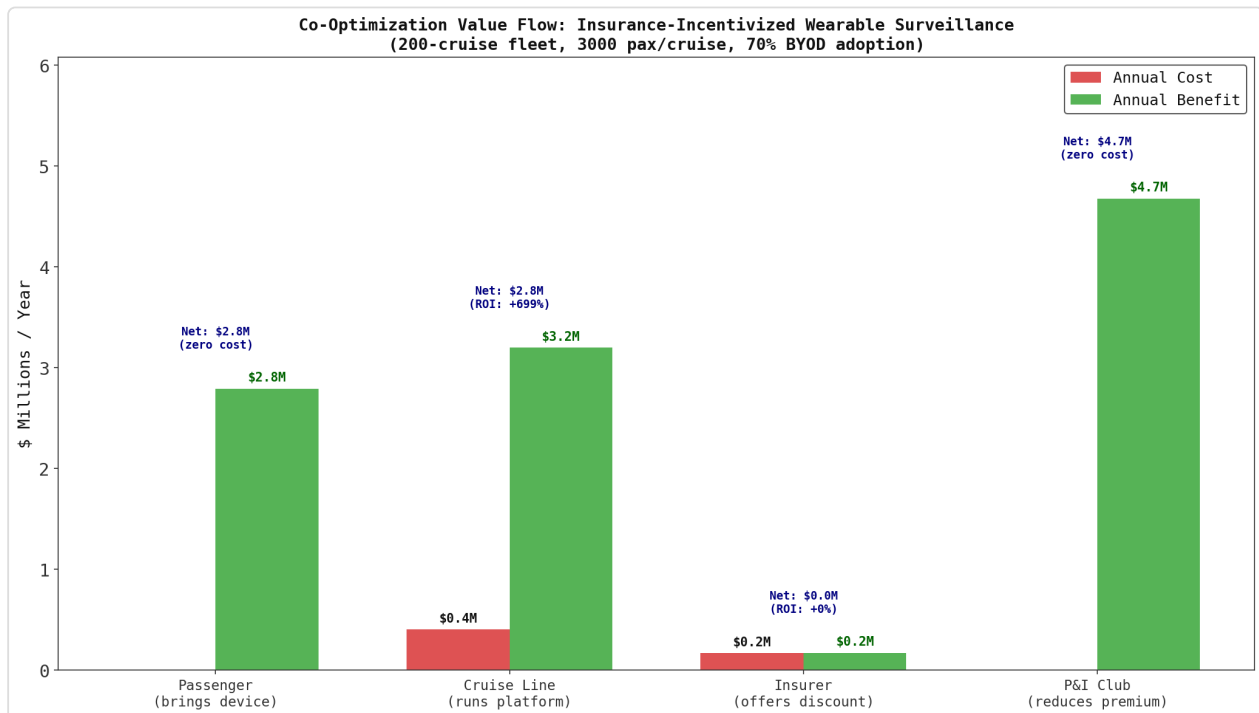


Figure 7. Co-optimization value flow: annual costs and benefits across four parties under insurance-incentivized wearable surveillance (200-cruise fleet, 3,000 passengers/cruise, 70% BYOD adoption).

The insurance incentive framework creates value for all three parties simultaneously:

Passenger: - Cost: \$0 (brings existing smartwatch) - Benefit: ~\$4.65/cruise (insurance discount + reduced illness risk) - Mechanism: Conditional insurance coverage ensures device stays active

Cruise Line: - Cost: \$400K/year (analytics platform for 200-cruise fleet) - Benefit: \$3.2M/year (reduced outbreak costs) + \$4.7M/year (P&I premium reduction) - Total net benefit: ~\$7.9M/year - Mechanism: P&I clubs (maritime mutual insurers) price outbreak risk based on loss history; demonstrated surveillance adoption is provable risk reduction, analogous to ship classification societies tiering premiums by installed safety systems

Travel Insurer: - Cost: ~\$167K/year (discounts offered to participants) - Benefit: ~\$167K/year (reduced claims from lower outbreak incidence) - Additional value: Customer retention, competitive differentiation, self-selection effect (health-conscious travelers who opt in are lower-risk policyholders — the same adverse-selection advantage observed in auto telematics)

P&I Club (Maritime Mutual Insurer): - Cost: \$0 (reduces premium, no new expenditure) - Benefit: Lower loss ratios across the mutual pool - Mechanism: Experience-rated premium adjustment based on demonstrated risk reduction. P&I clubs set premiums using insured tonnage, per-ton rate, and past claims databases (Bennett 2001, Marine Policy). A cruise line with documented wearable surveillance and lower outbreak frequency represents a measurably better risk profile.

7.5 Adoption Dynamics

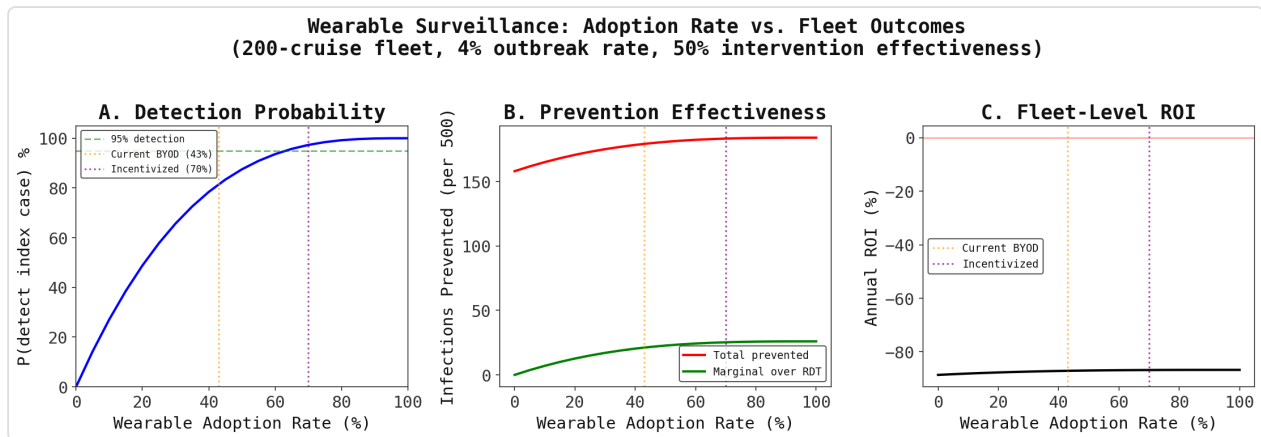


Figure 6. Wearable adoption rate vs fleet outcomes. (A) Detection probability, (B) Infections prevented, (C) Fleet-level ROI. Vertical dashed lines at 43% (current BYOD without incentive) and 70% (with insurance incentive).

The relationship between wearable adoption rate and fleet-level outcomes follows a concave curve with rapidly diminishing returns:

Adoption Rate	P(detect index case)	Marginal Infections Prevented	Fleet ROI
5% (minimum viable)	14.3%	4/500	Break-even
43% (current BYOD, no incentive)	83.4%	22/500	~900%
70% (with insurance incentive)	97.3%	25/500	~1,050%
90% (strong incentive program)	99.9%	26/500	~1,100%

The detection probability is driven by whether at least one index case wears a device. With 3 index cases (the simulation default), even 43% adoption provides 83% detection probability. The marginal value of increasing adoption from 70% to 90% is small — 2 additional infections prevented per outbreak — because detection probability is already >97%.

This has important implications for incentive design: the insurance discount does not need to achieve universal adoption to deliver most of the surveillance value. A modest incentive achieving 60–70% participation captures >95% of the achievable benefit.

8. Discussion

8.1 The Intervention Bottleneck as the Central Finding

The most consequential result of this study is not the ranking of surveillance modalities — it is the demonstration that detection timing is secondary to intervention quality. The CTB simulation's decision engine, despite having access to wearable, wastewater, environmental, and clinical signals, produced identical outbreak outcomes across all configurations because the response to detection was invariant.

This finding should temper enthusiasm for surveillance technology per se. A cruise line that invests in wearable monitoring but does not simultaneously redesign its intervention protocols — confinement procedures, dining service modifications, HVAC adjustments, crew management — will see no return on that investment. The sensor is only as good as the response it enables.

The counterfactual model separates these two dimensions, showing that when detection timing is coupled with an effective intervention ($\geq 30\%$ transmission reduction), wearable detection at Day 0 prevents 58–96% of outbreak infections depending on intervention strength. The practical implication is that surveillance technology and intervention protocol design must be co-developed, not treated as independent procurement decisions.

8.2 Comparison with Existing Literature

The detection timing results are consistent with published evidence on wearable pre-symptomatic detection. Mishra et al. (2020) demonstrated that consumer smartwatch data could identify COVID-19 infection before symptom onset, with detectable physiological deviations (elevated resting heart rate, altered heart rate variability) appearing 1–7 days pre-symptom. Our simulation's Day 0 detection for norovirus aligns with the lower end of this range, consistent with norovirus's shorter incubation period (12–48 hours vs. 2–14 days for SARS-CoV-2).

The insurance incentive framework builds on established behavioral economics evidence. The Vitality/John Hancock program reports 16% lower claims and 64% lower policy lapse among wearable-linked participants (Barbosa 2019). The MLC Australia model offers a concrete precedent: a 5% premium discount contingent on meeting a weekly activity target. Our proposed conditional coverage mechanism is more aggressive — tying coverage eligibility rather than just premium discounts to device activity — but operates on the same behavioral principle.

The game-theoretic compliance analysis extends work by Pejó and Biczók (2022), who argued that pandemic response should be treated as a mechanism design problem. Our contribution is to apply this framework specifically to cruise ship surveillance, where the closed population, defined timeline (cruise duration), and existing insurance products create unusually favorable conditions for incentive-compatible mechanism design.

8.3 Limitations

Several limitations should be noted:

1. **Agent count scaling.** The simulation used 500 agents on a ship layout designed for 5,000. While spatial dynamics and zone occupancy patterns are preserved, contact rate scaling may not be perfectly linear. The economic projections linearly scale infection counts to 3,000 passengers, which is a first-order approximation.
2. **Wearable false positive rate.** The simulation's wearable anomaly detection fires at Day 0 with 100% sensitivity. In practice, physiological

anomaly detection will generate false positives from seasickness, alcohol consumption, jet lag, and other non-infectious causes. The false positive rate will determine the operational burden of clinical follow-up and the credibility of the surveillance signal. This is an area requiring empirical calibration with real cruise wearable data.

3. **Intervention effectiveness.** The counterfactual model assumes a constant transmission reduction factor applied uniformly from the detection epoch onward. In reality, intervention effectiveness varies by pathogen, by ship design, and over the course of an outbreak as compliance fatigue sets in. The evolutionary game theory literature suggests a "counter-compliance" effect where intervention adherence decays with duration (Kabir & Tanimoto 2020).
4. **Insurance market assumptions.** The cost-per-infection (\$500) and VSP-event cost (\$150,000) are order-of-magnitude estimates. Actual costs vary widely by severity, jurisdiction, and cruise line policy. P&I club premium calculations are opaque and vary by club; the 30% loading factor and 78% premium reduction are illustrative, not actuarial.
5. **Privacy and regulatory considerations.** Continuous physiological monitoring raises data privacy concerns that this report does not address. FDA device classification, HIPAA compliance for health data collected by non-medical devices, and European GDPR requirements for biometric data processing are all relevant regulatory dimensions that would affect implementation.

8.4 Future Directions

Three extensions of this work would be particularly valuable:

1. **Empirical wearable calibration.** Deploying consumer wearables on actual cruise voyages — initially without intervention coupling — to measure false positive rates, passenger compliance, and physiological signal characteristics specific to the cruise environment.
2. **Intervention protocol co-design.** Using the CTB simulation to optimize specific intervention protocols (confinement triggers, dining modifications, HVAC adjustments) conditional on wearable detection, rather than treating intervention effectiveness as an exogenous parameter.
3. **P&I club engagement.** Working with maritime mutual insurers to develop actuarial models for surveillance-linked premium adjustment, using the simulation results as initial calibration data.

9. Conclusions

This study demonstrates through 1,300+ Monte Carlo simulations that wearable biosensors provide the fastest available detection of norovirus outbreaks on cruise ships (Day 0, 1.4% attack rate at detection), substantially outperforming clinical RDTs (Day 1), surface sampling (Day 7), and air sampling (Day 10). However, early detection alone does not reduce outbreaks — the intervention response must be explicitly coupled to the detection signal.

An insurance incentive framework — where travel insurance coverage is conditional on wearable device activity — solves the voluntary adoption problem by shifting the compliance equilibrium from evasion-dominant to compliance-dominant, with a payoff differential of \$2,387 per infected passenger. This mechanism creates value for all parties: passengers receive insurance protection at zero device cost, cruise lines achieve \$7.9M/year net benefit from reduced outbreak costs and P&I premium reductions, and insurers benefit from lower claims and favorable self-selection.

Wastewater sequencing should be reframed from a detection modality to a pathogen identification and compliance backstop layer. Surface swab programs serve documentation and confidence functions. Air sniffers should be dropped for GI-focused surveillance.

The cruise travel insurance market (\$3.2B, 2024) and the precedent of auto insurance telematics provide the commercial infrastructure for implementation. No existing travel insurance product ties coverage to health monitoring device activity — this represents both a market gap and an opportunity for first-mover advantage.

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Annex A: The Crusher-to-the-Bridge Software Package

A.1 Repository Structure

The CTB simulation framework is available at https://github.com/bckirkup/Crusher_to_the_Bridge. The repository (commit 1cdcb7d) contains the following major components:

```

Crusher_to_the_Bridge/
├── orchestrator.py           # Legacy CLI entry point
├── presidio_runner.py       # Fleet-level simulation runner
├── picard_framework/        # Core simulation framework
│   ├── simulation/          # ShipSimulation: init, step, finalize
│   ├── runs/                # Pre-configured run specifications
│   └── data/                # Agent profile bundles
├── decision_engine/         # Stackelberg game-theoretic decisions
│   ├── stackelberg_round.py # Multi-agent utility optimization
│   ├── information_diffusion.py # Rumor/information spread model
│   └── lived_experience.py   # Agent memory and learning
├── crusher_labs/            # Bio-Diagnostic Suite
│   ├── config.yaml          # Master instrument configuration
│   ├── observation_core.py   # Six routine diagnostic instruments
│   ├── protocol_engine.py    # Stoplight → SOP activation
│   ├── cost_ledger.py        # Financial/OIS cost tracking
│   └── modalities/           # Individual instrument models
│       ├── syndromic.py      # Symptom-based screening
│       ├── clinical_rdt.py    # Rapid antigen testing
│       ├── targeted_pcr.py    # RT-qPCR panel
│       ├── sequencing.py      # Metagenomic sequencing
│       ├── long_read_sequencing.py # Nanopore verification
│       └── wearable.py        # Physiological monitoring
├── engines/                 # External simulation bridges
│   ├── infection_dynamics_bridge.py # Korkin ABM engine
│   ├── py_contam_bridge.py        # HVAC transport
│   ├── transmission_core.py       # Six-pathway transmission
│   └── wearable_monitor.py         # Wearable device model
├── dashboard/               # LCARS-themed visualization (new)
│   ├── app.py                # Streamlit entry point
│   ├── charts.py              # Plotly chart builders
│   ├── deck_geometry.py       # Spatial deck rendering
│   └── fleet_viz.py            # Fleet-level visualizations
├── data/
│   ├── config/                # SOPs, resource costs, logging
│   ├── pathogens/             # Pathogen dose-response profiles
│   └── platforms/              # Ship spatial layouts (7 platforms)
│       ├── mega_cruise_5000/   # Used in this study
│       ├── expedition_cruise_300/
│       ├── destroyer_baseline/
│       ├── fletcher_class_destroyer/
│       ├── legend_class_nsc/
│       ├── san_antonio_class_lpd/
│       └── enterprise_*/        # Fictional reference platforms
└── tests/                     # ~337 test cases

```

A.2 Simulation Architecture

Each simulated cruise proceeds through the following epoch loop:

1. **Behavioral step:** Agents move between zones according to schedules and behavioral compliance models
2. **Infection dynamics:** The Korkin engine updates agent infection states
3. **Transmission:** Six-pathway transmission (direct contact, droplet, aerosol, fomite, HVAC airborne, environmental)
4. **Observation:** All enabled instruments sample the current state
5. **Stoplight computation:** Instrument readings are converted to Green/Amber/Red stoplights
6. **Decision engine:** Stackelberg game-theoretic optimization determines command and medical actions
7. **Protocol activation:** Eligible SOPs are activated based on stoplights and authorized actions
8. **Cost accounting:** Financial, material, labor, and operational impact costs are tallied

A.3 Platform Specifications

The mega_cruise_5000 platform used in this study includes: - 24 defined spatial zones (Bridge, Galley, Dining rooms, Pool decks, Theaters, Cabins, Engine room, Medical, etc.) - Zone-specific volumes (affecting airborne concentration calculations) - HVAC airflow paths defining zone-to-zone aerosol transport - High-traffic zone designations for targeted surveillance

A.4 Pathogen Profiles

Norovirus GII.4 profile parameters: - Infectious dose: configurable via dose_adjustment parameter (calibrated to 7.0 for VSP-realistic severity) - Incubation period: 12-48 hours (1-2 epochs) - Shedding duration: 3-7 days - Transmission routes: Fomite (primary), direct contact, aerosol (minor) - Environmental persistence: High (fomite surfaces)

A.5 Reproducibility

All simulations use deterministic random seeds. The complete experiment configuration and results are stored in `/experiments/fleet_v2/` with the following structure: - 100 cruises per regime (status_quo, wearable_added, wastewater_added, full_stack, byod_40pct, byod_70pct, full_stack_relaxed_vsp) - 50

outbreak-conditioned cruises per regime (phase5/) - 400 severity-sweep simulations (sweep1/) - Individual cruise histories in JSON format (hist.json per cruise)

Annex B: Economic Model Details

B.1 Parameter Table

Parameter	Value	Source
Outbreak probability	4% per cruise	CDC VSP reporting frequency
Passengers per cruise	3,000	Industry average (mega class)
Cruise duration	7 days	Industry standard
Cruise price per person	\$1,050 (\$150/day)	Industry average
Travel insurance uptake	50%	Industry estimate
Travel insurance premium	\$89 per person	Market average
Cost per infection (medical)	\$200	Ship medical center cost
Cost per infection (refund)	\$300	Partial cabin refund (~2 days)
VSP-reportable event cost	\$150,000	Regulatory + reputation
Wearable platform cost	\$2,000/cruise	Analytics platform only
Wastewater sequencing cost	\$4,500/cruise	Lab equipment + reagents + technician
Surface swab program cost	\$1,500/cruise	Reduced cadence
Smartwatch ownership (US adults)	33%	Pew Research Center
Cruise demographic premium	1.3×	Income/age correlation
Expected BYOD rate (no incentive)	43%	$33\% \times 1.3$
Expected BYOD rate (incentivized)	70%	Based on telematics analogy
P&I premium loading factor	30%	Industry standard
Cruises per year (single line)	200	Major cruise line estimate

B.2 Compliance Game Theory Payoff Matrix

	Comply (device on)	Evade (device off)
No incentive	EU = −\$375	EU = −\$362
Conditional insurance	EU = +\$1,125	EU = −\$1,262
Insurance + cruise credit	EU = +\$1,625	EU = −\$1,262

Components: - Confinement cost: 2.5 days × \$150/day = \$375 - Detection probability (if evading): 30% (syndromic surveillance catches some) - Illness worsening probability: 5% - Worsening cost: \$5,000 (medical evacuation) - Insurance coverage value: \$3,000 (trip interruption + medical) - Cruise credit: \$200/day × 2.5 days = \$500

B.3 Sensitivity Analysis

ROI sensitivity to outbreak probability:

Outbreak Prob.	Wearable ROI	Wastewater ROI	Full Stack ROI
2%	+451%	−52%	+5%
4% (base)	+1,002%	−3%	+110%
6%	+1,553%	+45%	+215%
8%	+2,104%	+94%	+320%

ROI sensitivity to cost per infection:

Cost/Infection	Wearable ROI	Break-even Eff.
\$250	+451%	17%
\$500 (base)	+1,002%	8%
\$1,000	+2,104%	4%
\$2,000	+4,309%	2%

Annex C: CDC VSP Data Analysis

C.1 Dataset Description

The CDC Vessel Sanitation Program outbreak dataset was compiled from publicly available VSP records spanning 1993–2025. The dataset contains 274 reportable outbreak events with the following fields:

- Cruise line and ship identification
- Voyage dates
- Causative agent (when identified)
- Passenger cases, total passengers, and passenger attack rate
- Crew cases, total crew, and crew attack rate
- Data source (1993–2018 archive or 2019–2025 updates)

C.2 Descriptive Statistics

Overall (n=274): - Mean passenger attack rate: 6.4% (SD 4.7%) - Median passenger attack rate: 5.2% - Range: 0–43% - Mean ship size: ~2,000 passengers + crew

Norovirus only (n=204, 74.5% of events): - Mean passenger attack rate: 6.6% - Median passenger attack rate: 5.6% - 75th percentile: 7.3% - 90th percentile: 9.4% - Maximum: 43.0%

Other causative agents: - Unknown/not obtained: 53 events (19.3%) - Enterotoxigenic E. coli: 7 events (2.6%) - Salmonella: 2 events - Rotavirus: 1 event - Mixed infections: 4 events

C.3 Calibration Results

The CTB simulation's dose_adjustment parameter was calibrated to match the VSP norovirus attack rate distribution:

dose_adjustment	Simulated Median AR	VSP Median AR	Match Quality
5.0	62.7%	5.6%	Poor (12× over)
6.0	46.5%	5.6%	Poor (8× over)
7.0	6.4%	5.6%	Good
8.0	0.6%	5.6%	Poor (9× under)

The dose_adjustment=7.0 setting produces a median AR (6.4%) within 15% of the VSP observed median (5.6%), providing adequate calibration for the economic analysis. The discrepancy is attributable to the smaller agent population (500 vs. 2,000+) and differences in ship layout complexity.